



A 3D model encryption scheme based on a cascaded chaotic system

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ABSTRACT

With the birth of the metaverse, 3D models have received extensive attention, and the security of information transmission continues to be an important issue. In this paper, we propose a 3D model encryption method based on a 2D chaotic system constructed via the coupling of the logistic map and infinite collapse (2D-LAIC) and on semi-tensor product (STP) theory. In terms of Lyapunov exponents, NIST test results, bifurcation diagrams, etc., 2D-LAIC exhibits better dynamical behavior than classical chaotic systems. 2D-LAIC can generate an unpredictable keystream, which is highly suitable for cryptography. Therefore, we propose a new 3D model encryption algorithm based on 2D-LAIC, named 3DME-SC. For a 3D model of the floating-point data type, XOR and STP processing are applied to the integer part and fractional part, respectively, of the model to obtain a 3D ciphertext model. The keystream required for XOR and STP processing is generated by 2D-LAIC. The results of a detailed security analysis and a comparative experimental analysis show that 3DME-SC exhibits good performance and effectiveness.

(Code: <https://github.com/Gao5211996/3D-model-encryption>)

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1. Introduction

With the development of modern science and technology, humankind has entered the information age, and the security of information transmission continues to be an important issue [1–3]. Presently, numerous methods for image protection, such as watermarking methods, information hiding methods, encryption methods [4–7], and other technologies, have been proposed. Image encryption technology processes an original image into a messy noise-like image, and the receiver then uses the correct secret key (unique) to reconstruct the original image [8–11]. This technology is the most commonly utilized method [12–14] for image protection.

The emergence of chaotic systems has promoted the development of image encryption technology. Because chaos is sensitive to initial values, easy to generate, and unpredictable for a long time [15–19], it meets the needs of symmetric cryptography [20–25]. Therefore, image encryption algorithms combined with chaos offer good security. Since the first chaotic encryption algorithm was proposed by Robert Matthews [26], image encryption algorithms based on chaos have gradually become prevalent

[27–29]. In 2019, Zhu et al. proposed a two-dimensional logistic-modulated-sine-coupling-logistic chaotic map (LSMCL), employed it in an image encryption algorithm and verified the effectiveness of the algorithm through theoretical analysis and simulation experiments [30]. In 2021, Zhu et al. proposed a three-dimensional image scrambling operation and a new two-dimensional diffusion structure [31]. In 2022, the authors proposed a sinusoidal-polynomial composite chaotic system (SPCCS) and proved that it satisfies Devaney's definition of chaos. A new image encryption algorithm was proposed based on SPCCS, which showed good security performance [32].

According to the investigation of Ref. [33], 32% of image encryption algorithms are based on chaos theory [34]. Enayatifar et al. [12] and Mahmud et al. [23] used a one-dimensional logistic map to generate the keystream, and Zhou et al. [35] combined two one-dimensional chaotic systems for keystream generation. However, the trajectories of one-dimensional chaotic systems are simple, and it is easy to derive their initial conditions and trajectories [36–38]. In addition, due to the limited computational accuracy of the computer used for the generation process, chaotic degradation will occur after a sufficiently large number of pseudorandom numbers have been generated, and the generated keystream becomes predictable at this time [39,40]. Therefore, to increase the security of cryptosystems, it is necessary to propose hyperchaotic

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systems with more complex dynamics. High-dimensional chaotic systems have more complex structures and chaotic behaviors, but they also have high implementation costs and low efficiency in practical applications. Therefore, considering both performance and cost, two-dimensional chaotic systems are the better choice [33]. Based on the logistic map [41] and infinite collapse [42], a two-dimensional chaotic system (2D-LAIC) is proposed in this paper. Compared with a one-dimensional chaotic system, 2D-LAIC is hyperchaotic and thus produces more complicated states. This is beneficial because when a chaotic system is employed in cryptography, the chaos performance determines the security level of the cryptographic system.

In addition to designing systems with complex structures and good dynamic behavior, combining chaotic systems with other techniques is another way to improve the security of encryption [43]. Zhou et al. [3] selected the Hartley transform for keystream generation. Wang and Gao [1] used a Boolean network for this purpose. In Refs. [12,23], images were encoded and decoded in the form of DNA and RNA. Genetic algorithms [44], Latin squares [45], and compressive sensing (CS) theory [46] have also been employed in image encryption algorithms. Refs. [1,10,28,47] applied semi-tensor product (STP) theory in the diffusion phase of image encryption. The time complexity of the STP (Mn^2 , $N \geq n$) is better than that of traditional matrix multiplication (MN^2), and STP theory has achieved good results in image encryption. However, this method requires the passing of a large number of residual keys. Hence, large-scale image encryption is highly resource consuming. In Ref. [28], the residual key situation was improved, but there were still a small number of residual keys to be passed. In this paper, we use STP theory to address the fractional part of the 3D model. In the proposed 3DME-SC algorithm, there is no need to pass any residual keys because the final ciphertext can be stored in floating-point form.

Compared with 2D images, 3D models provide a more realistic experience and have a more substantial visual impact. 3D models have been utilized in fields such as animation, medicine, and the metaverse. In addition, some military maps are displayed as 3D models. Ensuring the secure transmission of 3D models is a challenging task. Although many 3D model encryption algorithms have been proposed [48,49,52–55], they have some deficiencies. Xu et al. selected the Arnold cat map to scramble 3D models [48], and Raj et al. also chose to use the Arnold cat map in their 3D model encryption algorithm [49]. However, the Arnold cat map is periodic. Hence, it can be repeated a limited number of times to obtain the original plaintext image [56]. In the 3D model encryption algorithms proposed by Wang et al. [52] and Sun [53], only the integer part was operated on, while the fractional part of the data was disregarded. The authors treated the input 3D model as a 2D image. The 3D encryption algorithm presented in Ref. [54] has a small key space and is not resistant to brute force attacks. Jin et al. [55] proposed a 3D model encryption algorithm using Lu chaotic mapping. This algorithm has no scrambling process, and its robustness is very poor. In contrast, our proposed 3DME-SC algorithm operates on not only the integer part but also the fractional part of the model to be encrypted. Experiments show that 3DME-SC exhibits good security characteristics, including a large key space and good robustness.

In summary, this work makes the following three major contributions:

1. A 3D model encryption algorithm (3DME-SC) is proposed.
2. A 2D chaotic system (2D-LAIC), which has good dynamic behavior, is proposed.
3. Simulation experiments show that 3DME-SC exhibits good security characteristics and effectiveness.

The remainder of this paper is structured as follows. The second section explains some of the techniques applied in this paper.

In the third section, we present our proposed 2D chaotic system and analyze its performance. In the fourth section, we describe the 3DME-SC encryption algorithm, including the processes of secret key generation, encryption, and decryption. The fifth section reports experiments conducted to evaluate 3DME-SC and compare it with existing algorithms. The sixth section concludes the paper.

2. Related works

This section introduces STP theory and the 3D models employed in 3DME-SC.

2.1. STP theory

The STP operation, which was proposed by Cheng et al. [57], is a generalization of traditional matrix multiplication. This operation expands the range of matrices that can be multiplied while maintaining the essential properties of traditional multiplication [57]. Because the STP is reversible and meets the needs of symmetric cryptosystems, STP theory has been utilized in image encryption and has achieved good results [1,10,28,47]. However, STP theory has not yet contributed to the development of 3D model encryption algorithms. The STP is defined by combining the Kronecker product and traditional matrix multiplication.

Let $P \in \mathcal{M}_{a \times b}$ and $Q \in \mathcal{M}_{c \times d}$; the definition of the STP is

$$P \propto Q = (P \otimes I_{n/b})(Q \otimes I_{n/c}), \quad (1)$$

where n is the least common multiple of b and c , I is the identity matrix, “ $\propto$ ” is the STP, and “ $\otimes$ ” is the Kronecker product.

Example 1. There are matrices $P \in \mathcal{M}_{2 \times 2}$ and $Q \in \mathcal{M}_{3 \times 3}$,

$$T = P \propto Q = (P \otimes I_{6/2})(Q \otimes I_{6/3})$$

$$\begin{aligned} &= \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \propto \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 2 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 2 \\ 2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 2 \\ 2 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 2 & 1 \\ 2 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 1 & 2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 4 & 1 \end{bmatrix}. \end{aligned}$$

Example 2. A reversible example.

$$\begin{aligned} P &= T \cdot (Q \otimes I_2)^{-1} = \begin{bmatrix} 4 & 5 & 7 & 8 \\ 2 & 2 & 3 & 3 \\ 2 & 5 & 3 & 7 \end{bmatrix} \\ &\cdot \left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 4 & 5 & 7 & 8 \\ 2 & 2 & 3 & 3 \\ 2 & 5 & 3 & 7 \end{bmatrix} \\ &\cdot \begin{bmatrix} 2 & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 3 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 1 & 2 \end{bmatrix}. \end{aligned}$$

In the diffusion stage of 3DME-SC, we use the STP to diffuse the fractional part of the 3D model. Due to the reversible nature of the STP, the decryption algorithm is simply the inverse of the encryption algorithm.

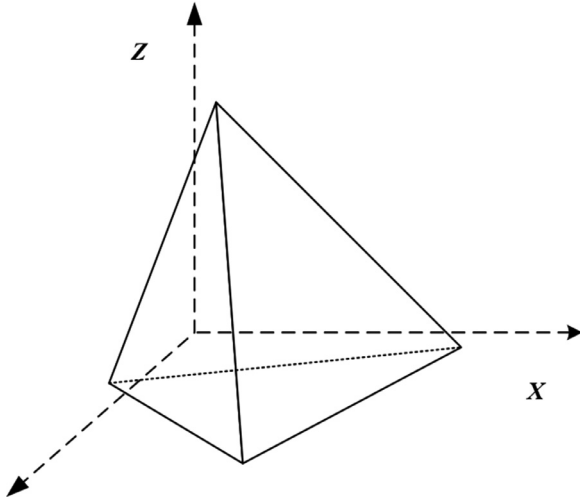


Fig. 1. A 3D model of STL.

Table 1
Information of 3D models.

Number	Name	Size	Length X	Length Y	Length Z
1	Banana	90000 × 3	137.7721	226.4492	82.0174
2	Clover	241872 × 3	48.0555	48.0555	40.5674
3	Hummer	270486 × 3	53.0500	31.1380	20.9390
4	MazdaRX-7	252894 × 3	109.7890	47.7316	31.9990
5	Pumpkin	90804 × 3	58.8330	62.1150	42.7500
6	Racingcar	87606 × 3	7.1720	16.3000	3.9060
7	Shamrock	111156 × 3	48.3205	42.5239	43.8946
8	Tank	72984 × 3	41.1720	23.3060	22.5020

2.2. 3D models

This paper proposes an encryption algorithm for 3D models stored in the STL file format. An STL file is a triangular representation of a 3-dimensional surface geometry that describes only the surface geometry of 3D objects and does not have any of the other attributes of other standard 3D models, such as color. STL files can only be used to represent closed faces or volumes. A 3D model defined in the STL file format is shown in Fig. 1.

The 3D model depicted in Fig. 1 consists of 4 triangles and 4 vertices. In the three-dimensional coordinate system, each vertex can be represented by a triple. In detail, the 3D model in Fig. 1 can be represented as $P = \{P(i)\}_{i=1}^4$, $P(i) = (x_{i,1}, y_{i,1}, z_{i,1})$, where $x \in R$, $y \in R$, and $z \in R$. The 3D models utilized in this paper are shown in Fig. 2. They are public 3D models (<https://www.3dzyk.cn/>). Table 1 lists information about these models, such as their names and the numbers of included triangles.

3. 2D-LAIC

In this section, the 2D-LAIC chaotic system is described, and its performance is evaluated in terms of certain evaluation indicators.

3.1. Definition of 2D-LAIC

A one-dimensional chaotic system is low in cost and simple in structure. However, these systems are easily predictable and do not exhibit hyperchaotic behavior [36–38]. Hyperchaotic dynamics are more complex than chaotic dynamics. In this paper, two one-dimensional chaotic systems are generalized into a two-dimensional chaotic system. The two selected chaotic systems are the logistic map and infinite collapse.

The mathematical definition of the logistic map is

$$w_{i+1} = \kappa \cdot w_i \cdot (1 - w_i), \quad (2)$$

where $\kappa \in [3.57, 4]$ is an adjustable parameter. The bifurcation diagram of the logistic map is shown in Fig. 3a. Within the interval throughout which κ can be adjusted, not all of the generated sequences are chaotic; there are some periodic sequences. Such sequences are not secure and thus should not be used in cryptographic systems.

The mathematical definition of infinite collapse is

$$v_{i+1} = \sin\left(\frac{\mu}{v_i}\right), \quad (3)$$

where μ is an adjustable parameter. The bifurcation diagram of infinite collapse is shown in Fig. 3b. Likewise, within the adjustable interval of μ , many periodic windows appear.

Referring to the structures of the logistic map and infinite collapse, we propose 2D-LAIC. The definition of 2D-LAIC is provided as follows.

Definition 1. The mathematical definition of 2D-LAIC ($f_{2D-LAIC}(x_i, y_i, \xi)$) is

$$\begin{aligned} f_{2D-LAIC}(x_i, y_i, \xi) \\ = \begin{cases} x_{i+1} = \sin(\pi \cdot \xi \cdot x_i \cdot (1 - x_i) + \pi \cdot \sin(\frac{\xi \cdot (1 - \xi)}{y_i}) + e) \\ y_{i+1} = \sin(\pi \cdot (1 - \xi) \cdot y_i \cdot (1 - y_i) + \pi \cdot \sin(\frac{x_i}{\xi}) + e) \end{cases} \end{aligned} \quad (4)$$

where ξ ($\xi \in (0, +\infty)$) is an adjustable parameter, e is the natural constant, and $X = \{x(i)\}_{i=1}^{\infty}$ and $Y = \{y(i)\}_{i=1}^{\infty}$ are the outputs of 2D-LAIC. When $\xi \in [4, +\infty)$, 2D-LAIC is hyperchaotic, meaning that 2D-LAIC can generate pseudorandom outputs that are more chaotic than those generated by non-hyperchaotic systems.

3.2. Lyapunov exponents

Lyapunov exponents (LEs) describe the degree of divergence of sequences obtained with similar initial values. The LEs of 2D-LAIC are shown in Fig. 4a. When $\xi \in [4, +\infty)$, 2D-LAIC always has two positive LEs. In this parameter region, 2D-LAIC is in a hyperchaotic state. The bifurcation diagrams shown in Fig. 4b and c verify this result. Because hyperchaotic states are more complex than chaotic states, 2D-LAIC is well suited for generating the keystream for a cryptosystem.

Furthermore, the effect of the natural constant e on 2D-LAIC is explained on the basis of comparative experiments conducted by replacing the natural constant e with values of 0 and 2.5 and analyzing the bifurcation diagrams and LEs of the resulting systems, as shown in Figs. 5 and 6.

When the constant e in 2D-LAIC is replaced with 0, corresponding to approximately $\xi = 4.7$, the system exhibits a periodic state. When e is replaced with 2.5, the system exhibits a periodic state of approximately $\xi = 4.3$. When the system state is periodic, the resulting sequence is ordered and thus is not suitable for use in a cryptographic system. Therefore, to avoid using the part of the parameter space of 2D-LAIC that can result in a periodic state in our proposed cryptosystem, we choose the natural constant e to ensure that 2D-LAIC will have better dynamic behavior.

3.3. NIST testing

The keystream used in a cryptosystem should be random. Otherwise, leakage of the secret key will occur. The National Institute of Standards and Technology (NIST) provides a standard suite for testing the randomness of data, expressed in terms of 15 indicators. The randomness of the x and y sequences in 2D-LAIC was tested following this standard, and the NIST test results are shown

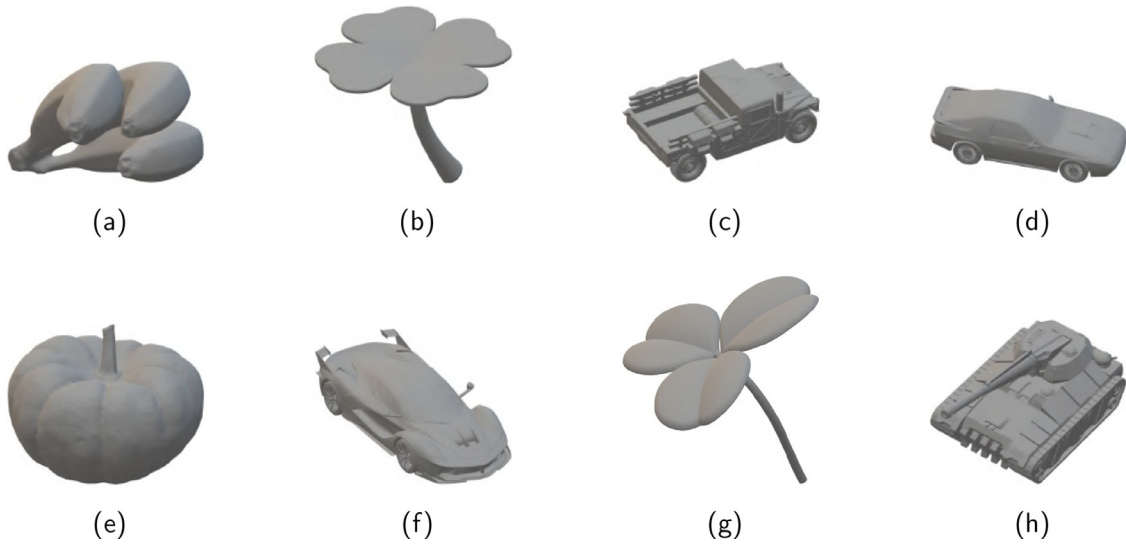


Fig. 2. 3D models.(a) Banana. (b) Clover. (c) Hummer. (d) MazdaRX-7. (e) Pumpkin. (f) Racingcar. (g) Shamrock. (h) Tank.

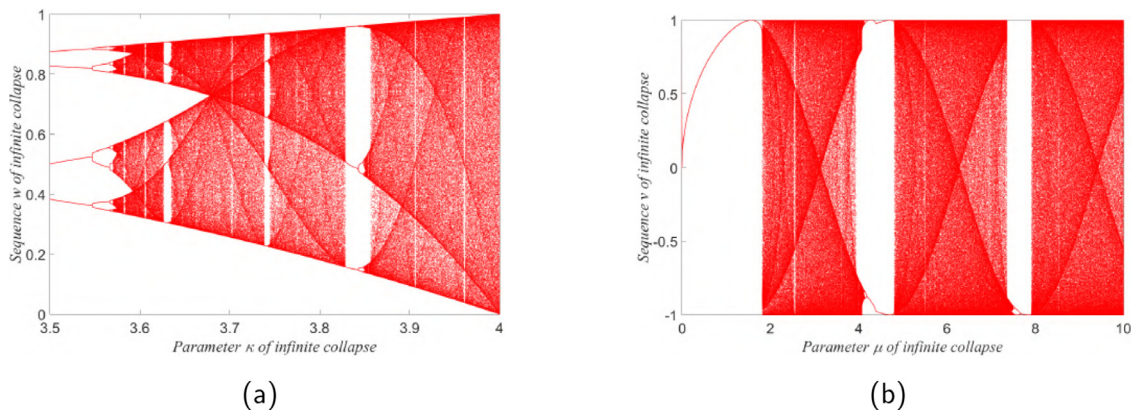


Fig. 3. Bifurcation diagram. (a) Bifurcation diagram of logistic map. (b) Bifurcation diagram of infinite collapse.

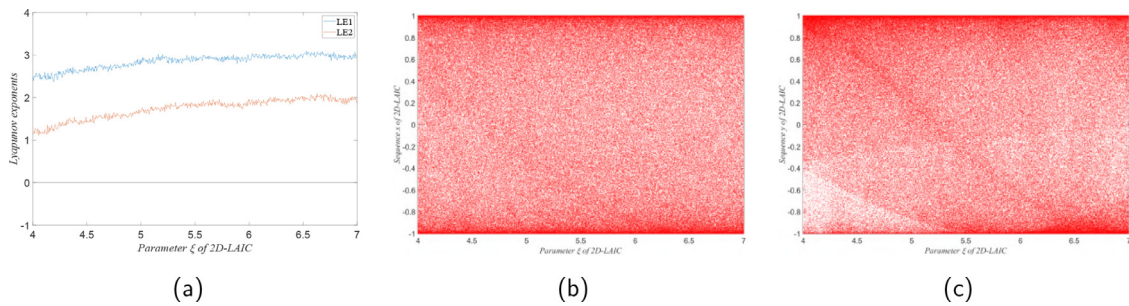


Fig. 4. Lyapunov exponents and bifurcation diagram of the 2D-LAIC. (a) LEs. (b) Bifurcation of x . (c) Bifurcation of y .

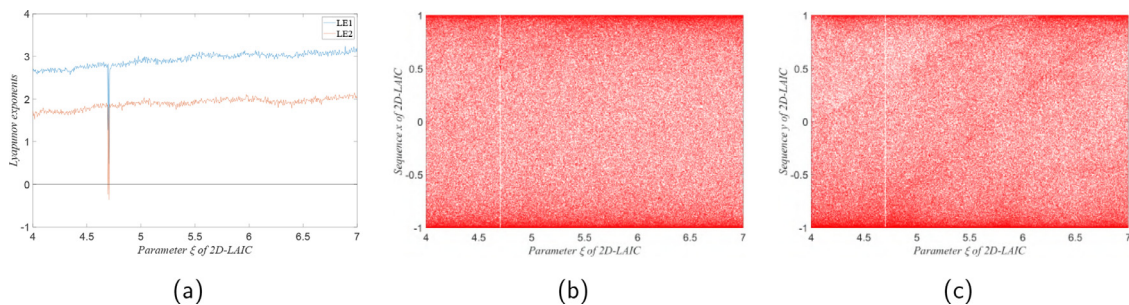


Fig. 5. Lyapunov exponents and bifurcation diagram of the 2D-LAIC with the natural constant e is replaced by 0. (a) LEs. (b) Bifurcation of x . (c) Bifurcation of y .

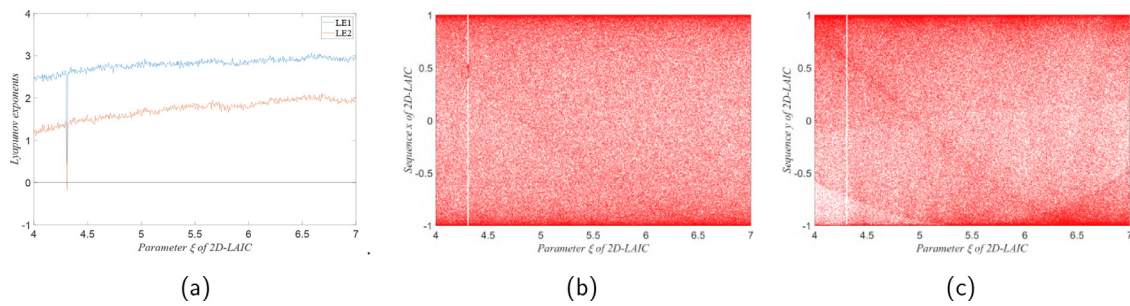


Fig. 6. Lyapunov exponents and bifurcation diagram of the 2D-LAIC with the natural constant e is replaced by 2.5. (a) LEs. (b) Bifurcation of x . (c) Bifurcation of y .

Table 2
NIST of the 2D-LAIC.

Number	Statistical test	x		y	
		P-value	Pass/Fail	P-value	Pass/Fail
1	Frequency	0.941144	Pass	0.311542	Pass
2	Block Frequency	0.941144	Pass	0.834308	Pass
3	Cumulative Sums	0.834308	Pass	0.739918	Pass
4	Runs	0.689019	Pass	0.057146	Pass
5	Longest Run	0.911413	Pass	0.213309	Pass
6	Rank	0.739918	Pass	0.875539	Pass
7	FFT	0.875539	Pass	0.834308	Pass
8	Non-Overlapping Template	0.788728	Pass	0.637119	Pass
9	Overlapping Template	0.834308	Pass	0.534146	Pass
10	Universal	0.875539	Pass	0.105618	Pass
11	Approximate Entropy	0.141256	Pass	0.242986	Pass
12	Random Excursions	0.637119	Pass	0.468595	Pass
13	Random Excursions Variant	0.534146	Pass	0.534146	Pass
14	Serial	0.534146	Pass	0.437274	Pass
15	Linear Complexity	0.534146	Pass	0.637119	Pass

Table 3
Symbol descriptions.

Symbol	Explain	Symbol	Explain
P	a 3D model	SP_I	integer part of SP
$x(x_1, x_2, x_3, \dots, x_n)$	X coordinate of 3D model	SP_D	fractional part of SP
$y(y_1, y_2, y_3, \dots, y_n)$	Y coordinate of 3D model	Z	keystreams is produced by U
$z(z_1, z_2, z_3, \dots, z_n)$	Z coordinate of 3D model	CD	ciphertext of fractional part
h	value of hash which is 256 bits	D	keystreams is produced by V
$(xa, ya, \xi a), (xb, yb, \xi b)$	initial value and parameters of 2D-LAIC	E	keystreams is produced by Y
m	the minimum value in P	CI_1	first ciphertext of integer part
M	the maximum value in P	CI_2	second ciphertext of integer part
X, Y, U, V	they are keystreams	C	ciphertext of 3D model
SP	scrambled matrix by Scramble P		

in Table 2. The sequences generated by 2D-LAIC exhibit good pseudorandomness, as seen from the fact that the results pass the test criteria in terms of all indicators.

4. Encryption algorithm for 3D models

3DME-SC is divided into three modules: key generation, an encryption algorithm, and a decryption algorithm.

4.1. Symbol descriptions

Table 3 defines some notations used to describe the cryptosystem.

4.2. Generation of secret keys using a hash function

The calculation of a hash function is a one-way (irreversible) process, and the hash value can be calculated at fairly high speed. Thus, hash functions are often used for secret key generation in cryptographic systems. In 3DME-SC, the secret key is generated using Hash256, and the input to the hash function is a 3D model.

The symbol P ($P \in \mathcal{M}_{n \times 3}$) represents a 3D model, expressed as $P \in \{(x_i, y_i, z_i) | i = 1, 2, 3, \dots, n\}$, where x , y , and z are floating-point values. The key generation algorithm of 3DME-SC is shown in Algorithm 1.

The secret keys xa , xb , ya , yb , ξa , and ξb are generated by Algorithm 1, which enables 2D-LAIC to generate a chaotic keystream. This keystream is used in the encryption phase of the cryptosystem. A schematic of Algorithm 1 is shown in Fig. 7. In Algorithm 1, ξa and ξb are the control parameters of 2D-LAIC. To ensure the security of the cryptosystem, the keystream should have good randomness. In 2D-LAIC, when $\xi \in [4, +\infty)$, the system exhibits characteristics of global chaos. Therefore, the desirable ranges of ξa and ξb are both $[4, +\infty)$. In this paper, for convenience of calculation, we choose $\xi a \in [4, 100]$ and $\xi b \in [4, 100]$.

4.3. Encryption algorithm

The encryption algorithm is divided into two stages: scrambling and diffusion. The diffusion stage is further divided into integer diffusion and fractional diffusion.

Algorithm 1: Key generation algorithm.

Input: P (P is 3D Model)
Output: $xa, xb, ya, yb, \xi a, \xi b$

```

1  $h \leftarrow \text{hash}(P, \text{SHA256})$   $\parallel h$  is 256 bits;
2 for  $i \leftarrow 1$  to 16 do
3    $d(i) \leftarrow h(16 \times (i-1) + 1 : 16 \times i)$ 
4 end
5 for  $i \leftarrow 1$  to 8 do
6    $q(i) \leftarrow d(2 \times i - 1) \oplus d(2 \times i) \parallel \oplus$  is XOR
7 end
8 for  $i \leftarrow 1$  to 6 do
9    $k(i) \leftarrow q(i) \oplus q(i+1) \oplus q(i+2)$ 
10 end
11 for  $i \leftarrow 7$  to 8 do
12    $k(i) \leftarrow k(i-1) \oplus k(i-6)$ 
13 end
14  $xa \leftarrow k(13)/2^{16}, xb \leftarrow k(14)/2^{16}, ya \leftarrow k(15)/2^{16}, yb \leftarrow k(16)/2^{16},$ 
    $\xi a \leftarrow k(17)/2^{16} \times 96 + 4, \xi b \leftarrow k(18)/2^{16} \times 96 + 4$ 

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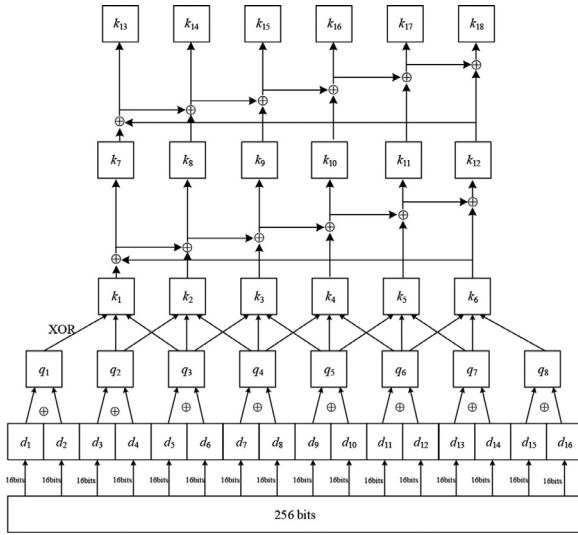


Fig. 7. The schematic diagram of Algorithm 1.

Input: $xa, xb, ya, yb, \xi a$, and ξb .

Step 1: Obtain the minimum value m ($m = \min(P)$) in P , and translate P to the positive axis of the coordinate axis,

$$P = \begin{cases} P & \text{if } m > 0 \\ P - m + 1 & \text{if } m \leq 0 \end{cases} \quad (5)$$

Step 2: Obtain the maximum value M ($M = \max(P)$) in the updated P ; accordingly, the ciphertext can be mapped to the interval $[0, A]$, $A = \{\min(2^i - 1) | 2^i \geq M\}$. M is the secret key for decryption.

Step 3: Generate the keystreams of the cryptosystem from 2D-LAIC based on the secret keys $xa, xb, ya, yb, \xi a$, and ξb . Discard the values from the first 200 iterations and generate chaotic keystreams X, Y, U , and V as follows: $[X, Y] = f_{2D-LAIC}(xa, ya, \xi a)$ and $[U, V] = f_{2D-LAIC}(xb, yb, \xi b)$, where xa, xb, ya, yb are the initial values of 2D-LAIC and ξa , and ξb are the parameters of 2D-LAIC.

Step 4: Transform P into $P \in \mathcal{M}_{n \times 3} \rightarrow P \in \mathcal{M}_{\frac{n}{3} \times 9}$ (when generating 3D models, ensure that $\text{mod}(n, 3) = 0$) and generate two sorting matrices, H^1 and H^2 . H^1 represents the sorted position coordinates of the sequence $X(1 : \frac{n}{3})$, and H^2 represents the position coordinates after the sequence $Y(1 : 9)$ is sorted.

Example 3.

$$X = [0.13, 0.25, 0.1, 0.36, 0.21, 0.96, 0.23, 0.89, 0.11],$$

$$X^1 = [0.1, 0.11, 0.13, 0.21, 0.23, 0.25, 0.36, 0.89, 0.96],$$

$$H = [3, 9, 1, 5, 7, 2, 4, 8, 6].$$

Step 5: Scramble P to obtain the scrambled matrix $SP = \{SP(j, i)\}_{j=1, i=1}^{n/3, 9}$ via

$$SP(j, i) = P(H^1(j), H^2(i)), i = 1, 2, 3, \dots, 9, j = 1, 2, 3, \dots, \frac{n}{3}. \quad (6)$$

Step 6: Extract the integer and fractional parts of SP to be separately diffused. Integer part: $SPI = \lfloor SP \rfloor$. Fractional part: $SPD = SP - \lfloor SP \rfloor$.

Step 7: Apply STP diffusion to the fractional part as follows:

$$\begin{aligned} Z &= U(9 \times i - 8 : 9 \times i), Z \in \mathcal{M}_{1 \times 9} \rightarrow Z \in \mathcal{M}_{3 \times 3}, \\ CD(N_2 \times i - N_2 + 1 : N_2 \times i, 1 : 9) &= SPD(N_2 \times i - N_2 + 1 : N_2 \times i, 1 : 9) \odot Z, i = 1, 2, 3, \dots, N_1, \\ Z &= U(9 \times N_1 - 8 : 9 \times N_1), Z \in \mathcal{M}_{1 \times 9} \rightarrow Z \in \mathcal{M}_{3 \times 3}, \\ CD(N_2 \times N_1 + 1 : N_2 \times N_1 + 1 : 9) &= SPD(N_2 \times N_1 + 1 : N_2 \times N_1 + 1 : 9) \odot Z. \end{aligned} \quad (7)$$

where N_2 is the adaptive parameter, and $N_1 = \lfloor n/3 \times N_2 \rfloor$. The ciphertext of the fractional part is $CD = CD / (\lfloor \max(CD) \rfloor + 1)$. $k_1 = \lfloor \max(CD) \rfloor + 1$ is the secret key, which is required for decryption.

Step 8: Apply XOR diffusion to the integer part in two steps: forward diffusion and reverse diffusion. Perform forward diffusion to obtain the first ciphertext matrix $Cl_1 = \{Cl_1(i, j)\}_{i=1, j=1}^{n/3, 9}$ via $Cl_1 = G_{fd}(V, A, SPI)$. Perform reverse diffusion to obtain the second ciphertext matrix $Cl_2 = \{Cl_2(i, j)\}_{i=1, j=1}^{n/3, 9}$ via $Cl_2 = G_{rd}(Y, A, Cl_1)$.

Definition 2. Definition of forward diffusion $Cl_1 = G_{fd}(V, A, SPI)$

Let $V \in \mathcal{M}_{1 \times 3n}$, $A \in \mathbb{R}^+$, and $SPI \in \mathcal{M}_{\frac{n}{3} \times 9}$.

- $\lfloor y \rfloor$ returns the nearest integer that is less than or equal to y . Obtain $D = \{D(1, i)\}_{i=1}^{3n}$ via

$$D(1, i) = \lfloor V(1, i) \times (A + 1) \rfloor, i = 1, 2, 3, \dots, 3n.$$

- “ $\rightarrow$ ” denotes the transformation of $P(m \times n)$ into $Q(s \times t)$, here, $m \times n = s \times t$. Obtain the new D via

$$D \in \mathcal{M}_{1 \times 3n} \rightarrow D \in \mathcal{M}_{\frac{n}{3} \times 9}.$$

- “ $\oplus$ ” denotes the XOR operation. Obtain $Cl_1(1, 1)$, $Cl_1 = \{Cl_1(i, 1)\}_{i=2}^{n/3}$, $Cl_1 = \{Cl_1(:, j)\}_{j=2}^9$ via

$$\begin{aligned} Cl_1(1, 1) &= SPI(1, 1) \oplus D(1, 1), \\ Cl_1(i, 1) &= Cl_1(i-1, 1) \oplus SPI(i, 1) \oplus D(i, 1), i = 2, 3, 4, \dots, \frac{n}{3}, \\ Cl_1(:, j) &= Cl_1(:, j-1) \oplus SPI(:, j) \oplus D(j, 1), j = 2, 3, 4, \dots, 9. \end{aligned}$$

Definition 3. Definition of reverse diffusion $Cl_2 = G_{rd}(Y, A, Cl_1)$

Let $Y \in \mathcal{M}_{1 \times 3n}$, $A \in \mathbb{R}^+$, and $Cl_1 \in \mathcal{M}_{\frac{n}{3} \times 9}$.

- Obtain $E = \{E(i, 1)\}_{i=1}^{3n}$ via $E(i) = \lfloor Y(i) \times (A + 1) \rfloor, i = 1, 2, 3, \dots, 3n$.
- Obtain the new E via $E \in \mathcal{M}_{1 \times 3n} \rightarrow E \in \mathcal{M}_{\frac{n}{3} \times 9}$.
- Obtain $Cl_2(1, 9)$, $Cl_2 = \{Cl_2(i, 9)\}_{i=2}^{n/3}$, and $Cl_2 = \{Cl_2(:, j)\}_{j=8}^1$ via

$$\begin{aligned} Cl_2(1, 9) &= Cl_1(1, 9) \oplus E(1, 9), \\ Cl_2(i, 9) &= Cl_2(i-1, 9) \oplus Cl_1(i, 9) \oplus E(i, 9), i = 2, 3, 4, \dots, \frac{n}{3}, \\ Cl_2(:, j) &= Cl_2(:, j+1) \oplus Cl_1(:, j) \oplus E(:, j), j = 8, 7, 6, \dots, 1. \end{aligned}$$

Step 9: The ciphertext is $C = Cl_2 + CD$, $C \in \mathcal{M}_{\frac{n}{3} \times 9} \rightarrow C \in \mathcal{M}_{n \times 3}$.

Output: C .

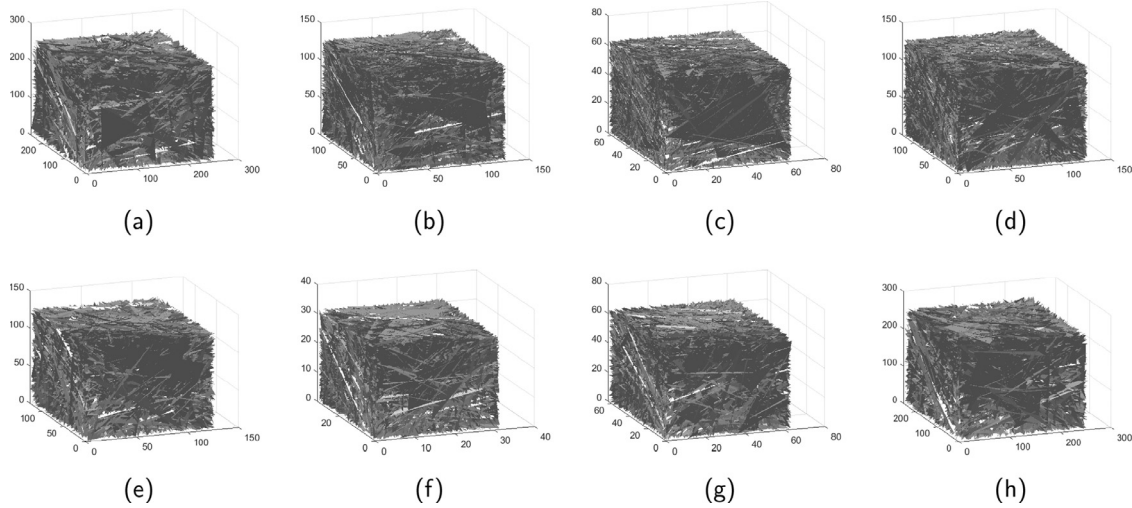


Fig. 8. Ciphred 3D models.(a) Ciphred Banana. (b) Ciphred Clover. (c) Ciphred Hummer. (d) Ciphred MazdaRX-7. (e) Ciphred Pumpkin. (f) Ciphred Racingcar. (g) Ciphred Shamrock. (h) Ciphred Tank.

4.4. Decryption algorithm

Theoretically, the decryption algorithm follows the inverse process of the encryption algorithm. However, due to the limited precision of real computers, errors will arise when a computer processes floating-point operations. Therefore, there will be some error between the decrypted model and the original model in practical applications. Nevertheless, this error is no greater than 10^{-13} , which is acceptable. The decryption algorithm is shown in Algorithm 2.

Algorithm 2: Decryption algorithm.

Input: A (Step 2), N_1 (Step 7), k_1 (Step 7), C (Step 9),
Output: P

```

1  $X, Y \leftarrow LAIC(xa, xb, \xi a)$  and  $U, V \leftarrow LAIC(xb, yb, \xi b)$   $C_{n \times 3} \leftarrow C_{n/3 \times 9}$ ,
 $Cl_2 \leftarrow \text{floor}(C)$ ,  $CD \leftarrow C - \text{floor}(C)$ ,  $CD \leftarrow CD \times k_1$   $E \leftarrow Y(1 : 3n)$ ,
 $E \leftarrow \text{floor}(E \times (A+1))$ ,  $E_{1 \times 3n} \leftarrow E_{n/3 \times 9}$ ,  $Cl_1(1, 9) \leftarrow Cl_2(1, 9) \oplus E(1, 9)$  for  $i \leftarrow$ 
2  $2$  to  $n/3$  do
3    $Cl_1(1, 9) \leftarrow Cl_2(i-1, 9) \oplus Cl_2(i, 9) \oplus E(i, 9)$ 
4 end
5 for  $i \leftarrow 8$  to  $1$  do
6    $Cl_1(:, 1) \leftarrow Cl_2(:, i+1) \oplus Cl_2(:, i) \oplus E(:, i)$ 
7 end
8  $D \leftarrow V(1 : 3N)$ ,  $D \leftarrow \text{floor}(D \times (A+1))$ ,  $D_{1 \times 3n} \leftarrow D_{n/3 \times 9}$ ,  $SPI(1, 1) \leftarrow$ 
 $Cl_1(1, 1) \oplus D(1, 1)$  for  $i \leftarrow 2$  to  $n/3$  do
9    $SPI(1, 9) \leftarrow Cl_1(i-1, 1) \oplus Cl_1(i, 1) \oplus D(i, 1)$ 
10 end
11 for  $i \leftarrow 2$  to  $9$  do
12    $SPI(:, i) \leftarrow Cl_2(:, i-1) \oplus Cl_2(:, i) \oplus D(:, i)$ 
13 end
14 for  $i \leftarrow 1$  to  $N_1$  do
15    $Z \leftarrow U(9 \times i - 9 : 9 \times i)$ ,  $Z_{1 \times 9} \leftarrow Z_{3 \times 3}$   $SPD(N_2 \times i - N_2 + 1 : N_2 \times i, 1 : 9) \leftarrow$ 
 $CD(N_2 \times i - N_2 + 1 : N_2 \times i, 1 : 9) \times \text{inv}(Z \oplus I_3)$ 
16 end
17  $Z \leftarrow U(9 \times N_1 - 8 : 9 \times N_1)$ ,  $Z_{1 \times 9} \leftarrow Z_{3 \times 3}$ 
 $SPD(N_2 \times N_1 + 1 : n/3, 1 : 9) \leftarrow CD(N_2 \times N_1 + 1 : n/3, 1 : 9) \times \text{inv}(Z \oplus I_3)$ 
 $H^1 \leftarrow X(1 : n/3)$ ,  $H^2 \leftarrow Y(1 : 9)$ ,  $SP \leftarrow SPD + SPI$  for  $i \leftarrow 1$  to  $9$  do
18   for  $j \leftarrow 1$  to  $n/3$  do
19      $P(H^1(j), H^2(i)) \leftarrow SP(j, i)$ 
20   end
21 end
22  $P_{n/3 \times 9} \leftarrow P_{n \times 3}$ 

```

5. Performance analysis

This section presents analyses of the security of 3DME-SC, including a key analysis, an information entropy analysis, a correlation analysis, an efficiency analysis, and a comparative analysis.

5.1. Simulation experiments

The encryption results for the models in Fig. 2 are shown in Fig. 8, and the decryption results are shown in Fig. 9. These simulation results show that 3DME-SC can complete the encryption and decryption of 3D models.

5.2. Information entropy analysis

The information entropy reflects the degree of confusion in information analysis. The calculation formula for this metric is [2]

$$H(k) = \sum_{i=0}^{2^L-1} p(k_i) \log_2 \frac{1}{p(k_i)}. \quad (8)$$

The information entropy can also verify whether the histogram distribution is uniform. The theoretical values of the information entropy are shown in Table 4. When the information entropy is closer to its theoretical value, the information distribution is more chaotic. The information entropy values of 3DME-SC for the models considered in this paper are shown in Table 5. Table 5 shows that the information distribution of the plaintext follows specific rules. However, the information distribution of the ciphertext obtained via 3DME-SC is uniform, and the information entropy reflected in the ciphertext is similar to the theoretical value. Therefore, the attacker cannot obtain valuable information from the ciphertext.

5.3. Statistical analysis

Statistical analysis includes histogram analysis and correlation analysis. An attacker will attempt to statistically identify the information distribution of the ciphertext and thus infer the information distribution of the plaintext. Therefore, the distribution of the ciphertext information should be random.

5.3.1. Histogram analysis

In 2D image encryption algorithms, the distributions of the plaintext and ciphertext pixel values are determined via histogram analysis. Likewise, histogram analysis can yield statistical coordinate value distributions in 3D model encryption algorithms. The plaintext histograms of the 3D models are shown in Fig. 10, and the ciphertext histograms are shown in Fig. 11. As shown in Figs. 10 and 11, the distributions of the plaintext histograms are

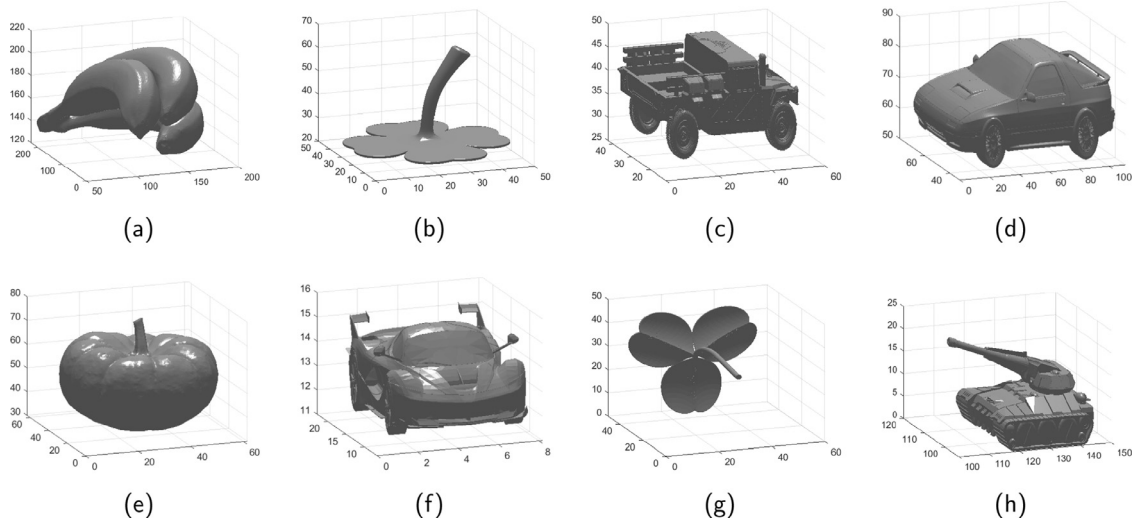


Fig. 9. Decrypted 3D models. (a) Decrypted Banana. (b) Decrypted Clover. (c) Decrypted Hummer. (d) Decrypted MazdaRX-7. (e) Decrypted Pumpkin. (f) Decrypted Racingcar. (g) Decrypted Shamrock. (h) Decrypted Tank.

Table 4
Theoretical value of information entropy.

L	1	2	3	4	5	6	7	8	9	10
2^L	1	4	8	16	32	64	128	256	512	1024
Information entropy	1	2	3	4	5	6	7	8	9	10

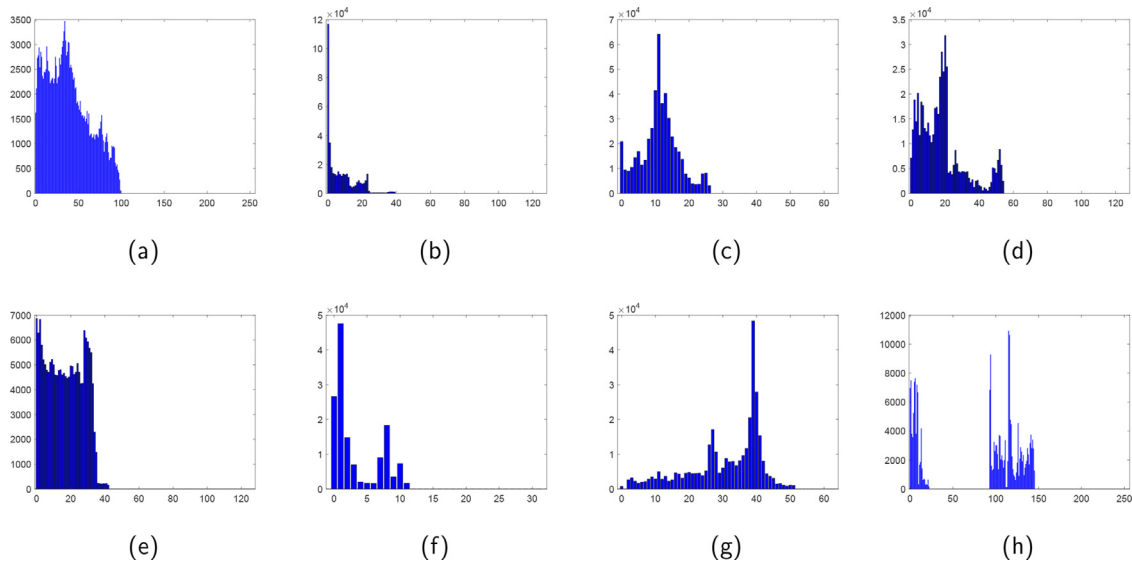


Fig. 10. Histogram analysis of original 3D model. (a) Banana. (b) Clover. (c) Hummer. (d) MazdaRX-7. (e) Pumpkin. (f) Racingcar. (g) Shamrock. (h) Tank.

Table 5
Information entropy of 3DME-SC.

3D models	Plaintext	Ciphertext	Theoretical value
Banana	4.7896	7.9994	8
Clover	2.6679	6.9999	7
Hummer	3.0425	5.9999	6
MazdaRX-7	3.8669	6.9999	7
Pumpkin	3.8017	6.9996	7
Racingcar	2.0057	4.9999	5
Shamrock	5.0131	5.9999	6
Tank	5.7986	7.9992	8

Table 6
Chi-square tests of 3DME-SC.

3D models	Plaintext	Ciphertext
Banana	2.8375×10^5	216.0289
Clover	3.0480×10^6	131.6032
Hummer	9.4887×10^5	68.1123
MazdaRX-7	1.1003×10^6	104.0117
Pumpkin	3.4295×10^5	144.9766
Racingcar	4.3496×10^5	35.6454
Shamrock	6.9594×10^5	45.6905
Tank	1.0359×10^6	231.7761

uneven, whereas the ciphertext histograms exhibit uniform distributions; therefore, an attacker cannot obtain valuable information from the ciphertext.

Table 6 shows the chi-square test results for 3DME-SC. These results show that the ciphertext value distributions obtained via

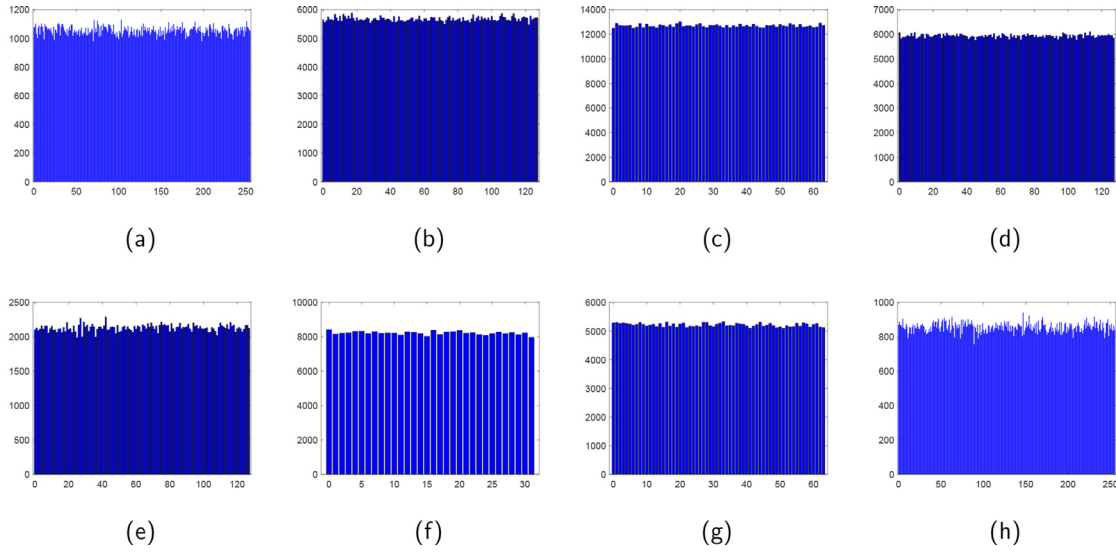


Fig. 11. Histogram analysis ciphered 3D models. (a) Ciphered Banana. (b) Ciphered Clover. (c) Ciphered Hummer. (d) Ciphered MazdaRX-7. (e) Ciphered Pumpkin. (f) Ciphered Racingcar. (g) Ciphered Shamrock. (h) Ciphered Tank.

Table 7
Correlation analysis.

3D models	Plaintext			Ciphertext		
	X direction	Y direction	Z direction	X direction	Y direction	Z direction
Banana	0.9849	0.9946	0.9691	-0.0047	0.0001	-0.0037
Clover	0.9969	0.9968	0.9997	-0.0005	-0.0028	-0.00008
Hummer	0.6978	0.7041	0.7215	0.0019	-0.0018	-0.0019
MazdaRX-7	0.9977	0.9918	0.9936	0.00009	0.0019	0.0026
Pumpkin	0.9691	0.9838	0.9690	0.0030	-0.0010	-0.0003
Racingcar	0.8887	0.9835	0.9639	-0.0010	-0.0038	-0.0021
Shamrock	0.9976	0.9951	0.9975	-0.0010	-0.0026	-0.0059
Tank	0.9631	0.9329	0.9898	-0.0017	0.0010	0.0037
#Average	-	-	-	-0.0005	-0.0011	-0.0009

3DME-SC are uniform, which means that 3DME-SC can resist statistical attacks.

5.3.2. Correlation analysis

For a secure algorithm, the correlation between adjacent coordinates in the ciphertext should be small. If there is substantial correlation between adjacent coordinates in the ciphertext, then an attacker will be able to infer the entire ciphertext based on only a few ciphertext coordinates, allowing an equivalent algorithm to be constructed. The calculation formula for the correlation is [2]

$$r_{uv} = \frac{\frac{1}{N} \sum_{i=1}^N (u_i - \frac{1}{N} \sum_{i=1}^N u_i)(v_i - \frac{1}{N} \sum_{i=1}^N v_i)}{\sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - \frac{1}{N} \sum_{i=1}^N v_i)^2} \sqrt{\frac{1}{N} \sum_{i=1}^N (u_i - \frac{1}{N} \sum_{i=1}^N u_i)^2}}. \quad (9)$$

If adjacent coordinates are strongly correlated, then r_{uv} will be near 1. If adjacent coordinates have a weak correlation, then r_{uv} will be near 0. The correlation analysis of 3DME-SC is shown in Table 7. It can be seen that adjacent coordinates in the plaintext have a strong correlation, whereas adjacent coordinates in the ciphertext obtained via 3DME-SC have a weak correlation, which further shows that 3DME-SC has a solid ability to resist statistical attacks. The “Shamrock” model is selected as an example to show a complete correlation analysis in Fig. 12. When there is a strong correlation between adjacent coordinates, the graph will appear aggregated. Otherwise, the graph will be divergent. Figure 12 verifies the results in Table 7: adjacent ciphertext coordinates are only weakly correlated.

5.4. Differential attack analysis

A strong encryption algorithm should be sensitive to the plaintext, which means that the ciphertext obtained after a small change in the plaintext should be very different from the original ciphertext. The number of pixels change rate (NPCR) and the unified average changing intensity (UACI) are indicators applied to test the difference between two ciphertext images. Their calculation formulas are [2]

$$NPCR = \frac{\sum_{i=1}^N \sum_{j=1}^3 D(i, j)}{3 \times N} \times 100\%, D(i, j) = \begin{cases} 0 & [C_1(i, j)] = [C_2(i, j)] \\ 1 & [C_1(i, j)] \neq [C_2(i, j)] \end{cases} \quad (10)$$

$$UACI = \frac{1}{3 \times N} \sum_{i=1}^N \sum_{j=1}^3 \frac{|[C_1(i, j)] - [C_2(i, j)]|}{2^L - 1} \times 100\%. \quad (11)$$

The formulas for calculating the mathematical expectations of the NPCR and UACI are

$$\begin{cases} NPCR_E = (1 - 2^{-L}) \times 100\% \\ UACI_E = \frac{1}{2^{2L}} \sum_{i=1}^{2^L-1} i(i+1) \times 100\% \end{cases} \quad (12)$$

The mathematical expectations of the NPCR and UACI are shown in Table 8.

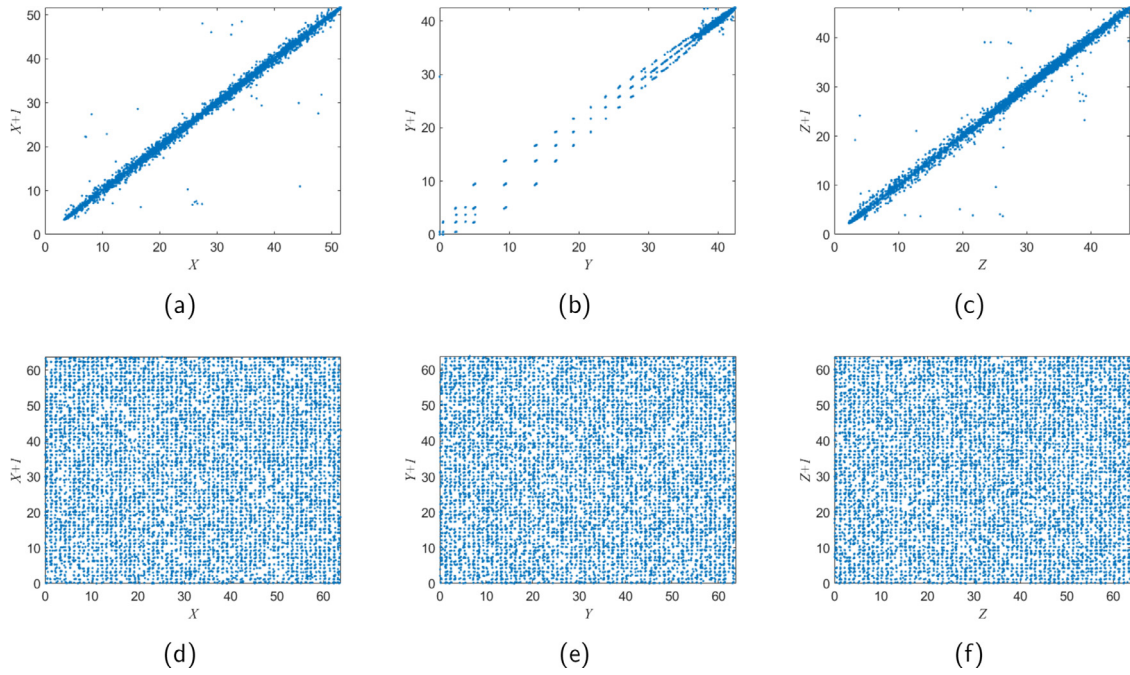


Fig. 12. Correlation analysis of Shamrock. (a) Original correlation of X and $X + 1$. (b) Original correlation of Y and $Y + 1$. (c) Original correlation of Z and $Z + 1$. (d) Ciphered correlation of X and $X + 1$. (e) Ciphered correlation of Y and $Y + 1$. (f) Ciphered correlation of Z and $Z + 1$.

Table 8

The mathematical expectations of NPCR and UACI.

L	5	6	7	8
$NPCR_E(\%)$	96.8750	98.4375	99.2188	99.6093
$UACI_E(\%)$	34.3750	33.8542	33.5938	33.4635

Table 9

UACI and NPCR of 3DME-SC.

3D models	L	NPCR(%)	Difference(%)	UACI(%)	Difference(%)
Banana	8	99.6082	-0.0011	33.4656	0.0021
Clover	7	99.2178	-0.001	33.6133	0.0195
Hummer	6	98.4474	0.0099	33.8535	-0.0007
MazdaRX-7	7	99.2187	-0.0001	33.6017	0.0079
Pumpkin	7	99.2248	0.0060	33.5860	-0.0078
Racingcar	5	96.8756	0.0006	34.3623	-0.0127
Shamrock	6	98.4397	0.0022	33.8507	-0.0035
Tank	8	99.6079	-0.0014	33.4642	0.0007

When an encryption algorithm's NPCR and UACI are closer to the theoretical values, the algorithm is more resistant to differential attacks. We randomly selected 50 points in the plaintext and added one to each to obtain a new ciphertext, which was then passed through the decryption algorithm of 3DME-SC. The average NPCR and UACI values of 3DME-SC are shown in Table 9. This differential attack analysis shows that 3DME-SC is sensitive to the plaintext, meaning that the ciphertext obtained after minor changes to the plaintext is very different from the original ciphertext, and that the NPCR and UACI indicators are similar to their theoretical values. Therefore, 3DME-SC has a strong ability to resist differential attacks.

5.5. Key analysis

A hash function is used for key generation in 3DME-SC. In addition, other parameters, such as A (Step 2), N_1 (Step 7), and k_1 (Step 7), are applied during decryption. Only the keys generated by

the hash function are considered. The keyspace of 3DME-SC is 2^{256} (greater than 2^{100}), which is sufficient to resist brute force attacks.

The way to verify that an encryption algorithm resists a chosen-plaintext attack is to check whether [50,51]

$$P_1(i, j) \oplus P_2(i, j) = C_1(i, j) \oplus C_2(i, j).$$

To verify that 3DME-SC has the ability to resist chosen-plaintext attacks, only the integer parts of the 3D model plaintext and ciphertext are considered. If the above equation holds, then the algorithm has poor resistance to chosen-plaintext attacks. The ability of 3DME-SC to resist chosen-plaintext attacks is evaluated as shown in Fig. 13.

The NPCR is used to evaluate the similarity between Fig. 13a and b and between Fig. 13c and d. The NPCR between Fig. 13a and b is 99.1654%, and that between Fig. 13c and d is 97.2412%. Thus, it is evident that $P_1(I, J) \oplus P_2(I, J) \neq C_1(I, J) \oplus C_2(I, J)$.

The input to the hash function is a 3D model. Due to the characteristics of hash functions, different plaintexts generate different secret keys. In a chosen-plaintext attack, the attacker can temporarily use the encryption and decryption systems, but the key is unknown. If different plaintexts generate the same key, then the attacker will be able to construct an equivalent key-cracking algorithm based on the cryptosystem. Therefore, 3DME-SC can resist chosen-plaintext attacks. Chosen-plaintext attacks are the most potent among the four classic attack types. Therefore, this analysis is sufficient to show that 3DME-SC can resist all four classic attack methods.

5.6. Robustness analysis

When ciphertexts are transmitted and stored over the Internet, noise and data loss will be present. Therefore, their effects on the decryption performance are analyzed in this section. The robustness analysis of 3DME-SC is shown in Figs. 14 and 15. This analysis shows that 3DME-SC exhibits good robustness; despite the occurrence of noise and data loss, the receiver can retrieve the approximate information of the plaintext through the decryption algorithm of 3DME-SC.

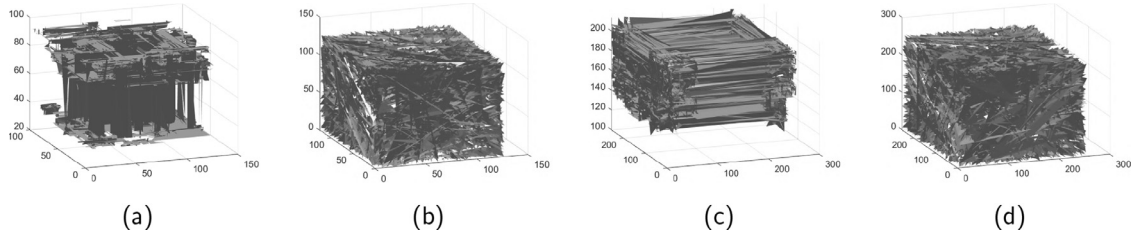


Fig. 13. Chosen-plaintext attacks. (a) Banana XOR Tank. (b) Ciphered Banana XOR Ciphered Tank. (c) Clover XOR MazdaRX-7. (d) Ciphered Clover XOR Ciphered MazdaRX-7.

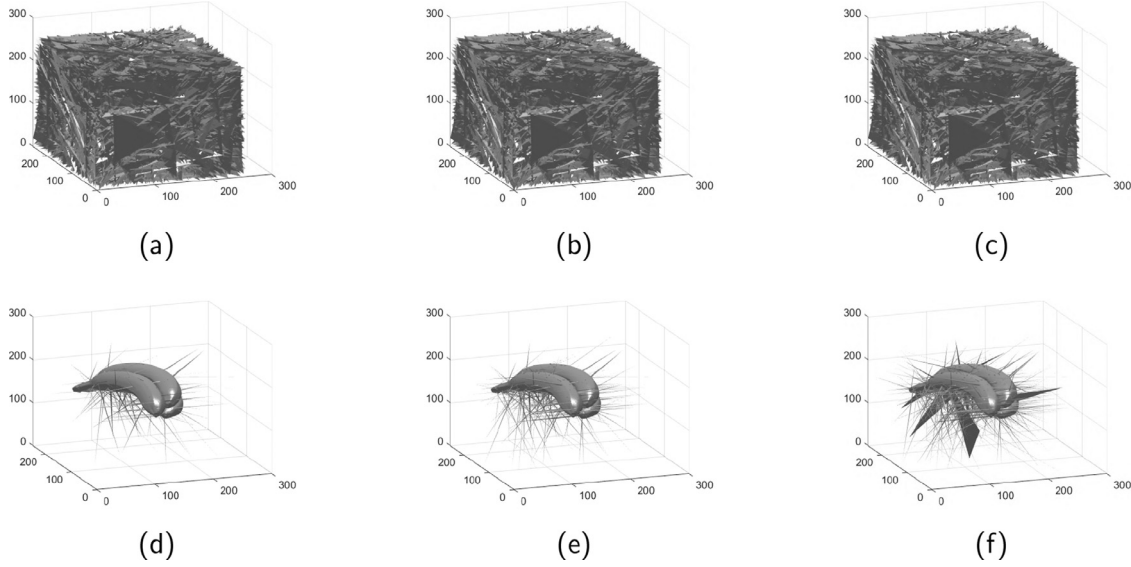


Fig. 14. Data loss of Banana. (a) Ciphered Banana of 50 coordinate lost ($C(1 : 50, :) = 0$). (b) Ciphered Banana of 100 coordinate lost ($C(1 : 100, :) = 0$). (c) Ciphered Banana of 200 coordinate lost ($C(1 : 200, :) = 0$). (d) Decrypted Banana of (a). (e) Decrypted Banana of (b). (f) Decrypted Banana of (c).

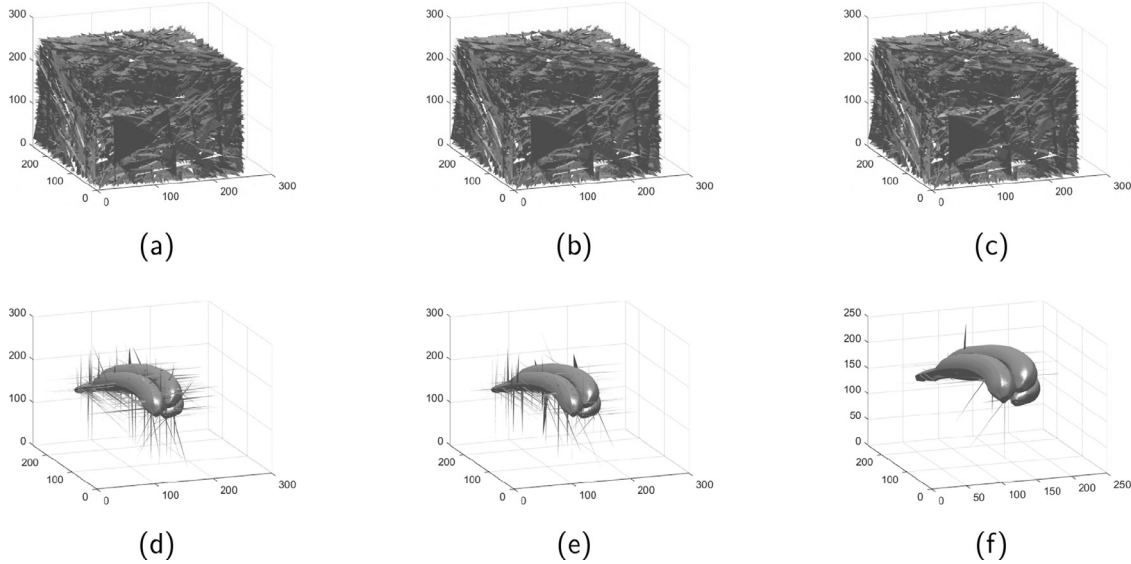


Fig. 15. Different noise attacks of Banana. (a) Ciphered Banana of 0.001 salt&pepper noise. (b) Ciphered Banana of 0.0005 salt&pepper noise. (c) Ciphered Banana of 0.0001 salt&pepper noise. (d) Decrypted Banana of (a). (e) Decrypted Banana of (b). (f) Decrypted Banana of (c).

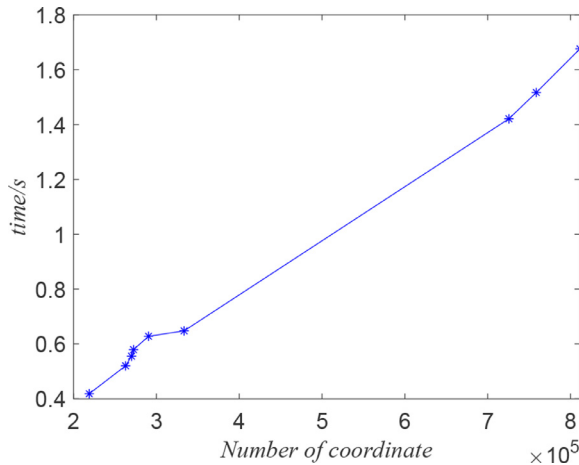
5.7. Efficiency analysis

The system environment is described as follows: the Windows 10 operating system, an i5-4210 CPU, and MATLAB R2019a. The encryption times of 3DME-SC for the tested models are shown in Table 10 and Fig. 16, and these results are compared with those

of existing 3D model encryption algorithms in Table 11. Compared with the data reported in Refs. [48,52,53,55,58], we processed larger-scale models when recording the run times of 3DME-SC. The comparison of the results shows that even when processing larger-scale data, the run time of 3DME-SC is still short. Therefore, 3DME-SC is more efficient.

Table 10
Efficiency analysis.

Number	Name	Number of coordinate	Time/s
1	Banana	270,000	0.628161
2	Clover	725,616	1.421181
3	Hummer	811,458	1.676305
4	MazdaRX-7	758,682	1.516879
5	Pumpkin	272,412	0.555534
6	Racingcar	262,818	0.530490
7	Shamrock	333,468	0.647891
8	Tank	218,952	0.418648



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